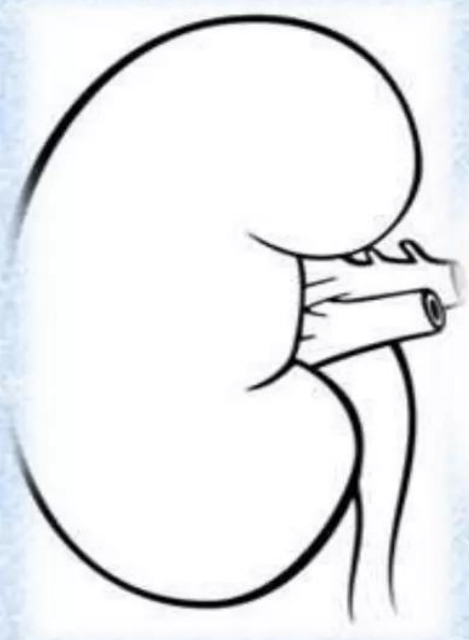




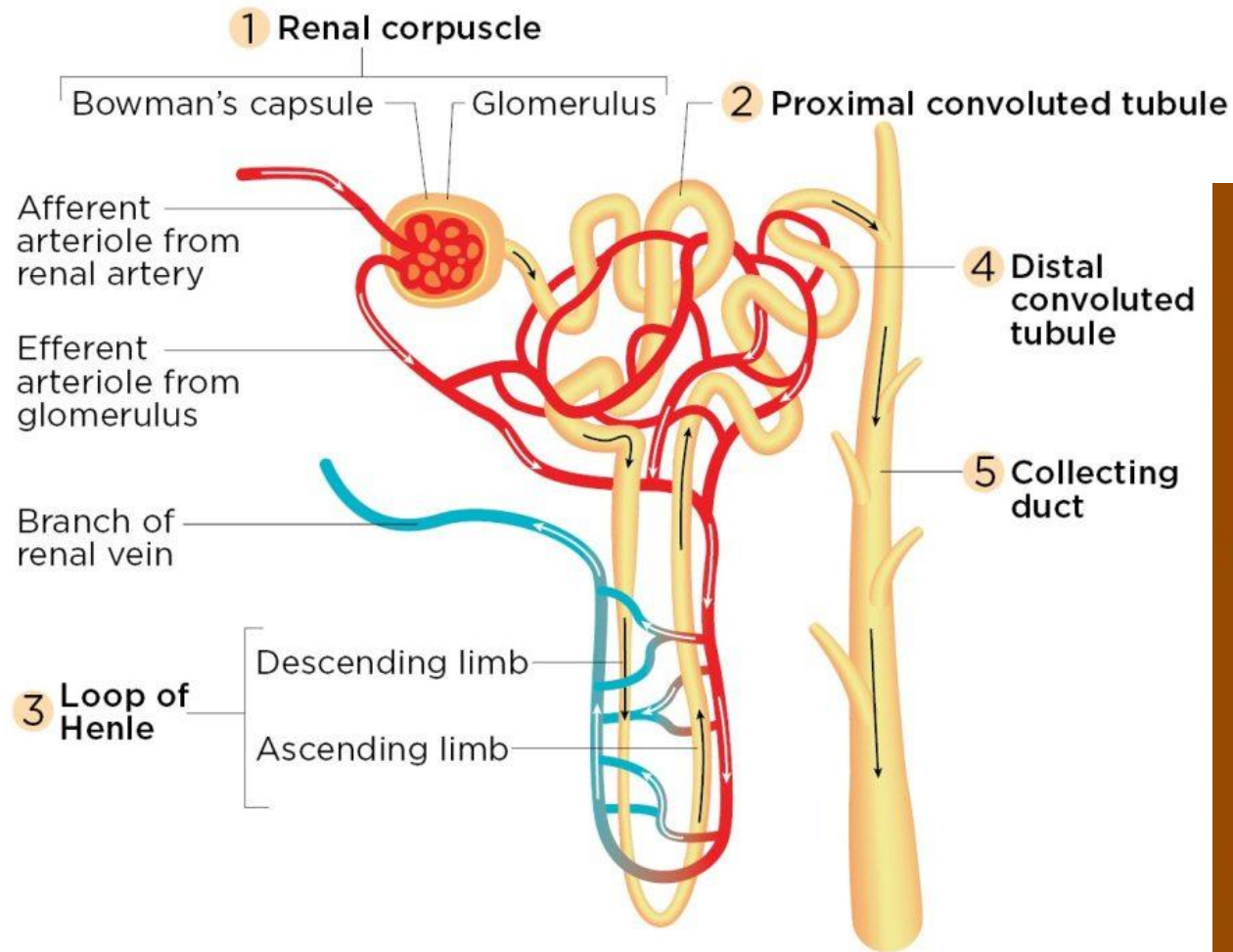
RENAL PHARMACOLOGY



By Firew A. B.Pharma, MSc

T.g. Address @HKO1336

Fig 3. **Anatomy of a nephron**



5 FUNCTIONS OF KIDNEY

- ✓ **Fluid Balance**
- ✓ **Waste Elimination**
- ✓ **PH regulation**
- ✓ **Blood pressure regulation**
- ✓ **Red blood cell production**

Common Kidney diseases

Disease	Description	Symptoms	Metabolism
Acute Kidney Injury (AKI)	Sudden loss of kidney function, often reversible.	Decreased urine output, swelling, fatigue	Reduced clearance of renally excreted drugs
Chronic Kidney Disease (CKD)	Progressive loss of kidney function over months or years.	Anemia, hypertension, bone disease	Altered pharmacokinetics, need for dose adjustments
Glomerulonephritis	Inflammation of the glomeruli, affecting filtration.	Hematuria, proteinuria, hypertension	Possible altered drug absorption and excretion
Nephrotic Syndrome	Increased glomerular permeability leading to significant protein loss in urine.	Edema, hypoalbuminemia, hyperlipidemia	Changes in drug binding and distribution
Polycystic Kidney Disease	Genetic disorder characterized by the growth of numerous cysts in the kidneys.	Pain, hematuria, kidney stones	Variable impact depending on cyst size and number
Diabetic Nephropathy	Kidney damage resulting from long-term diabetes.	Proteinuria, progressive kidney function loss	Requires careful monitoring of drug doses
Hypertensive Nephropathy	Kidney damage due to chronic high blood pressure.	Hypertension, proteinuria, CKD	Affects renal blood flow and drug clearance

Pharmacokinetics in Renal Impairment

1. **Absorption:** - Generally **Unchanged**
2. **Distribution:** - **Altered Protein Binding:** Reduced albumin levels in renal impairment can increase the free fraction of drugs, affecting their activity and toxicity
3. **Increased Volume of Distribution:** Edema and fluid retention can alter the volume of distribution for hydrophilic drugs.
4. **Metabolism:** - Decreased Metabolism:
 - Hepatic Metabolism: Some drugs may have altered hepatic metabolism due to changes in enzyme activity or hepatic blood flow.
5. **Excretion:-** Reduced Renal Clearance: - Accumulation Risk: Higher risk of drug accumulation and toxicity due to reduced excretion.
 - Dose Adjustments: - Necessary for Many Drugs
6. **Monitoring:** - Therapeutic Drug Monitoring (TDM): Essential for drugs with narrow therapeutic windows.
 - Renal Function Tests: Regular monitoring of serum creatinine, GFR, and electrolytes is crucial

Drug Classes in Renal Impairment

Drug Class	Examples	Primary Use in Renal Pharmacology
Diuretics	Furosemide, Hydrochlorothiazide, Spironolactone, Mannitol	Manage edema, hypertension, heart failure
Antihypertensives	Lisinopril, Losartan, Amlodipine, Metoprolol	Control blood pressure, protect renal function
Erythropoiesis-Stimulating Agents (ESAs)	Epoetin alfa, Darbepoetin alfa	Treat anemia in chronic kidney disease (CKD)
Phosphate Binders	Sevelamer, Calcium acetate, Lanthanum carbonate	Control hyperphosphatemia in CKD
Calcimimetics	Cinacalcet	Manage secondary hyperparathyroidism in CKD
Potassium Binders	Sodium polystyrene sulfonate, Patiromer	Treat hyperkalemia in CKD
Vitamin D Analogues	Calcitriol, Paricalcitol	Treat secondary hyperparathyroidism, manage bone mineral disorders
Immunosuppressants	Tacrolimus, Cyclosporine	Prevent rejection in kidney transplantation
Antimicrobials	Trimethoprim-sulfamethoxazole, Ciprofloxacin	Treat urinary tract infections, manage infections in renal patients
Analgesics (non-nephrotoxic)	Acetaminophen, Tramadol	Pain management in renal disease

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Diuretics Overview

Type of Diuretic	Examples	Site of Action	Mechanism of Action	Clinical Indications
Loop Diuretics	Furosemide, Bumetanide	Ascending loop of Henle	Inhibit $\text{Na}^+ \text{-} \text{K}^+ \text{-} 2\text{Cl}^-$ symporter, increasing excretion of Na^+ , K^+ , Cl^- , and water	Edema, heart failure, hypertension
Thiazide Diuretics	Hydrochlorothiazide, Chlorthalidone	Distal convoluted tubule	Inhibit $\text{Na}^+ \text{-} \text{Cl}^-$ symporter, reducing Na^+ reabsorption	Hypertension, mild edema
Potassium-Sparing Diuretics	Spironolactone, Eplerenone	Collecting ducts	Antagonize aldosterone, increasing Na^+ excretion while retaining K^+	Heart failure, hypertension, hyperaldosteronism
Osmotic Diuretics	Mannitol	Proximal tubule, descending loop of Henle	Increase osmotic pressure in the tubules, preventing water reabsorption	Acute renal failure, cerebral edema, glaucoma
Carbonic Anhydrase Inhibitors	Acetazolamide	Proximal convoluted tubule ↓	Inhibit carbonic anhydrase, reducing reabsorption of bicarbonate	Glaucoma, altitude sickness, metabolic alkalosis

Loop Diuretics

- Loop diuretics are a class of diuretics that act on the ascending loop of Henle in the nephron to promote the excretion of water and electrolytes.

Examples:

- - Furosemide (Lasix)
- - Bumetanide
- - Torsemide
- - Ethacrynic Acid

Mechanism of Action:

- - Inhibition of $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ Symporter:
- - Loop diuretics inhibit the $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ symporter in the thick ascending limb of the loop of Henle.
- - This inhibition leads to decreased reabsorption of sodium, potassium, and chloride ions.
- - As a result, water follows these electrolytes into the urine, increasing urine output.

Side Effects:

- - **Electrolyte Imbalance:** Hypokalemia, hyponatremia, hypomagnesemia.
- - **Dehydration:** Excessive diuresis can lead to volume depletion.
- - **Ototoxicity:** High doses or rapid intravenous administration can cause hearing loss.
- - **Hypotension:** Due to significant fluid loss and reduced blood volume.

Monitoring:

- - Electrolyte Levels: Regular monitoring of serum potassium, sodium, and magnesium.
- - Renal Function: Monitoring of serum creatinine and blood urea nitrogen (BUN) to assess kidney function.
- - Fluid Status: Monitoring of weight and signs of dehydration.

Diuretics

- Administered orally, IV & IM

❖ Therapeutic uses

- Generally **reserved** for situations that require rapid or massive mobilization of fluid.
 - A powerful drug
- Acute pulmonary edema associated with CHF
- Edema of hepatic, cardiac or renal origin
- Hypertension
- Anion overdose
- Acute hypercalcemia & hyperkalemia
- Acute renal failure

❖ Adverse effects

- » Hyponatremia, hypochloremia & dehydration
- » Postural hypotension
- » Hypokalemia
- » Ototoxicity
- » Hyperglycemia
- » Hyperuricemia
- » Teratogenic in experimental animals
- » Hyperlipidemia, hypomagnesemia & hypocalcemia
- » Allergic rxn

❖ Drug interactions

- » Digoxin
- » Ototoxic drugs
- » Potassium-sparing diuretics
- » Antihypertensive agents
- » NSADs

2. Thiazides and Related Diuretics/TDs/

2. Thiazides and Related Diuretics/TDs/

- Most commonly used diuretics
- Similar action to that of LDs but differ in:
 - Lower diuresis produced by TDs
 - Not used during renal insufficiency

❖ Classes of drugs

- » Hydrochlorothiazide
- » Chlorothiazide
- » Methyclothiazide
- » Chlorthalidone
- » Indapamide
- » Metolazone

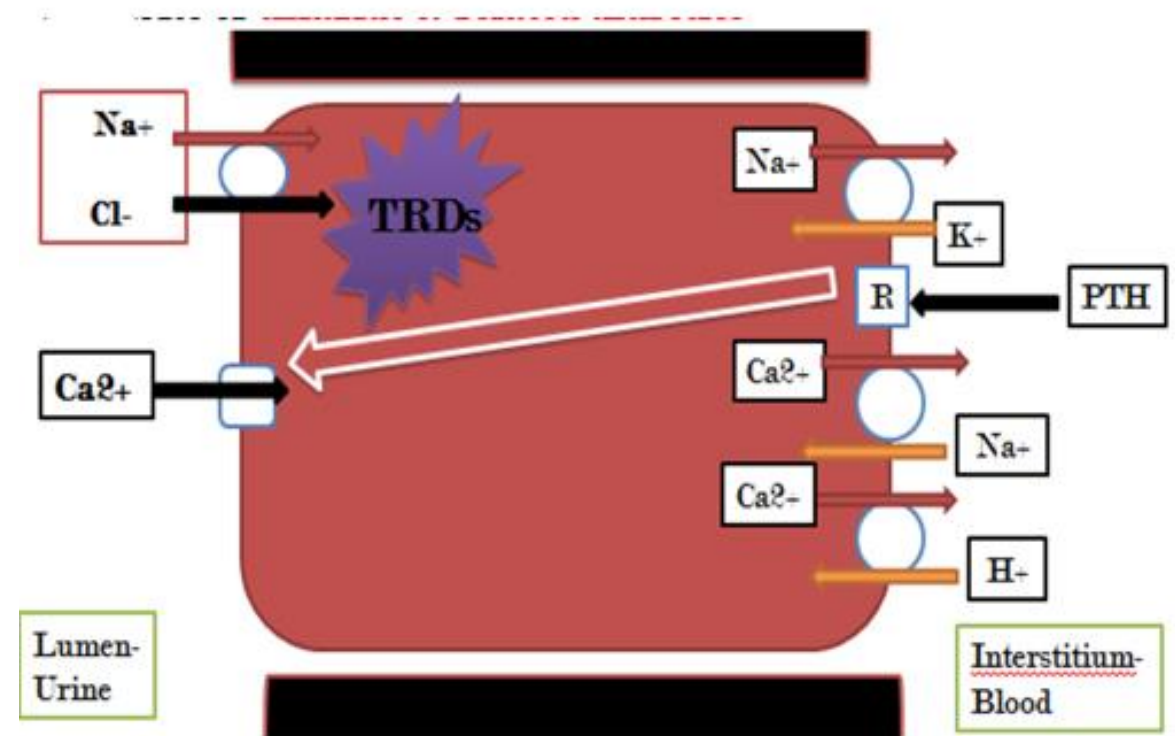
- Most of them given via PO

- All have equal maximum diuretic effects
- Sometimes called *ceiling diuretics*

❖ MOA

- Site of action is on luminal membrane
- Block **Na⁺/Cl⁻-cotransport** in the early segment of DCT
- Their action is dependent on adequate kidney function
 - Ineffective at low GFR (< 20 mL/min)

MOA



TDs...

❖ Therapeutic uses

1. Essential Hypertension
2. Edema- CHF, renal or hepatic Ds
3. Diabetes insipidus
4. Hypercalciuria

❖ Adverse effects

- Similar to that of LDs
- Exception
 - Lack ototoxicity
 - Hypercalemia

❖ Drug interactions

- Nearly identical to LDs
- Digoxin- cardiotoxicity
- K⁺-sparing diuretics- beneficial
- Antihypertensive drugs-augment
- NSAIDs- antagonizes TDs effect

3. Potassium-Sparing Diuretics/PSDs/

- Modest diuresis
- Hyperkalemia
 - Used to counteract TDs & LDs-mediated K⁺ loss

• Two groups:

I. Aldosterone antagonists

- Spironolactone & eplerenone

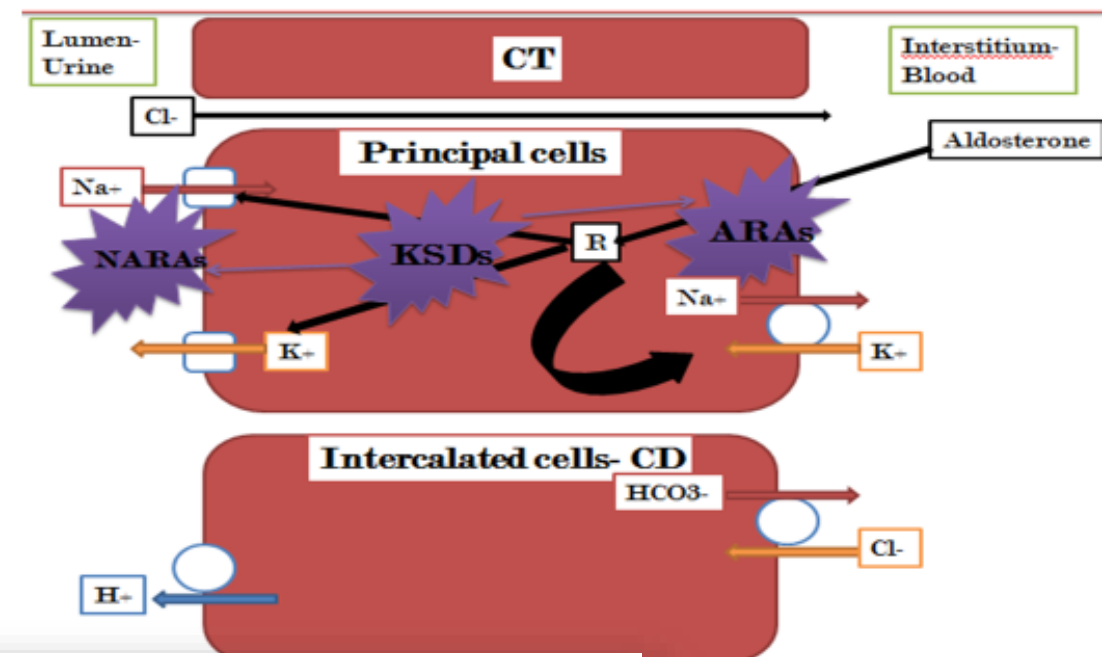
❖ MOA

- Inhibits the aldosterone-mediated reabsorption of Na⁺ & K⁺ secretion [Na⁺-K⁺ exchange]
- Blocks the actions of aldosterone in the distal nephron.
- Have delayed onset of action (up to two days). Why?

Therapeutic uses

- Hypertension & edema
- Heart failure
- Hyperaldosteronism

MOA



❖ Adverse effects

1. Hyperkalemia- Rx-insulin inj
2. Benign & malignant tumors-thyroid, testes, mammary and liver
3. Endocrine effects- Gynecomastia, amenorrhea, impotence & hirsutism

❖ Drug interactions

1. Thiazide & loop diuretics
2. ACEIs, ARBs & RIs

PSDs...

2. Non-aldosterone Antagonists

- Triamterene & amiloride

❖ MOA

- Block Na⁺ channels
 - Disrupts sodium-potassium exchange in the distal nephron
 - Direct inhibitor of the exchange mechanism
 - Have quick onset of action

❖ Therapeutic uses

- Hypertension and edema
- Combined with other diuretics

❖ Adverse effects

- Hyperkalemia, nausea & vomiting, dizziness & blood dyscrasias

4. Osmotic Diuretics/ODs/

- Four compounds
 - Mannitol, urea, glycerin, isosorbide
 - Hydrophilic cpds
- Used in
 - Acute renal failure
 - Intracranial pressure(ICP) & IOP
- ❖ Osmotic Diuretic possesses the ff four properties:
 - Freely filtered at glomerulus
 - Undergoes minimal reabsorption
 - Not metabolized to a significant degree
 - Pharmacologically inert

- Mannitol
- MOA
 - Promotes diuresis
 - By creating an osmotic force within nephron lumen
 - PCT, thin descending limb & collecting duct
 - Inhibits passive reabsorption of water
 - Given by IV
- Therapeutic uses
 - Prophylaxis of renal failure
 - Reduction of ICP
 - Reduction of IOP
- Adverse effects
 - Edema
 - Headache
 - Nausea & vomiting
 - Fluid and electrolyte imbalance

5. Carbonic Anhydrase Inhibitors/CAIs/

❖ Acetazolamide

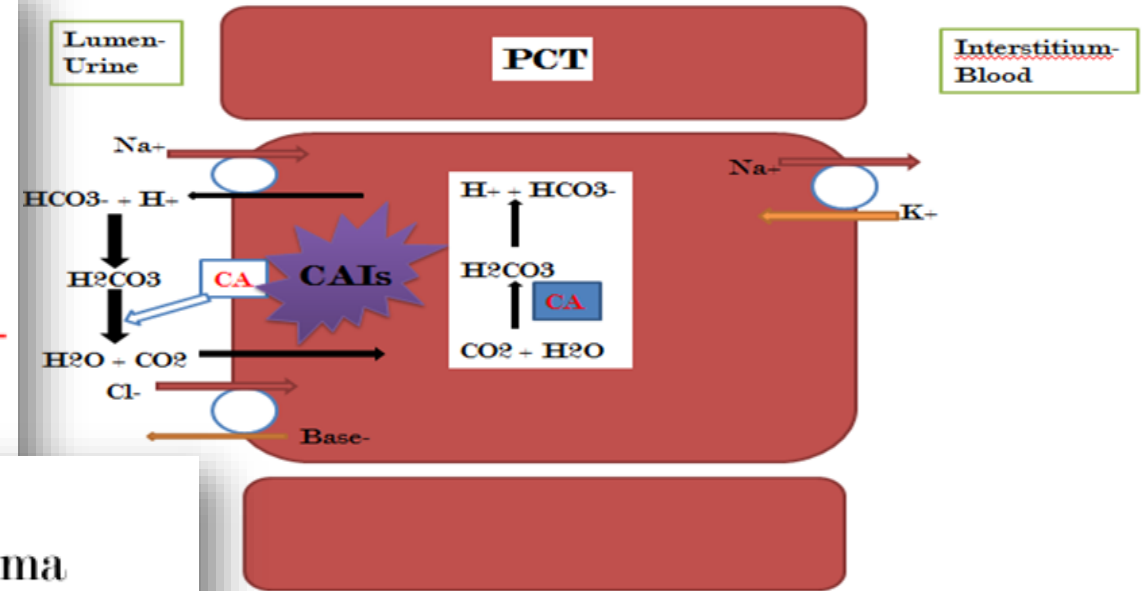
- Inhibits the enzyme CA in the luminal memb of PCT
 - $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightarrow \text{H}^+ + \text{HCO}_3^-$
 - Decrease exchange of **Na⁺ for H⁺**
 - Increased urinary excretion of **Na⁺, K⁺ & HCO₃⁻**
- Inhibitors block NaHCO₃ reabsorption and cause diuresis
 - Much less efficacious
 - Mild diuresis

▪ Therapeutic uses

1. Open-angle glaucoma
2. Urinary alkalization
3. Metabolic alkalosis

▪ Adverse effects

1. Metabolic acidosis
2. Potassium depletion
3. Renal stone formation
4. Drowsiness
5. Paresthesia- skin sensation



Antihypertensives in Renal Disease

Class of Antihypertensive	Examples	Mechanism of Action	Indications in Renal Disease
ACE Inhibitors	Enalapril, Lisinopril, Ramipril	Inhibit angiotensin-converting enzyme (ACE), blocking conversion of angiotensin I to angiotensin II	Chronic kidney disease (CKD), proteinuria, diabetic nephropathy
Angiotensin II Receptor Blockers (ARBs)	Losartan, Irbesartan, Valsartan	Block angiotensin II receptors, preventing vasoconstriction and aldosterone release	Hypertension, proteinuria, CKD progression, diabetic nephropathy
Calcium Channel Blockers (CCBs)	Amlodipine, Diltiazem, Verapamil	Block calcium channels in vascular smooth muscle cells, leading to vasodilation	Hypertension, proteinuria, CKD progression, renal artery stenosis
Beta Blockers	Atenolol, Metoprolol, Carvedilol	Block beta-adrenergic receptors, reducing heart rate and contractility	Hypertension, heart failure, CKD with co-existing cardiovascular disease
Diuretics	Hydrochlorothiazide, Furosemide	Promote diuresis, reducing blood volume and blood pressure	Hypertension, CKD with volume overload

Comparison of ACE Inhibitors and ARBs

Aspect	ACE Inhibitors	ARBs
Mechanism of Action	Inhibit angiotensin-converting enzyme (ACE), preventing the conversion of angiotensin I to angiotensin II.	Block angiotensin II receptors, preventing the binding of angiotensin II to its receptors.
Effect on Angiotensin II	Decrease circulating levels of angiotensin II.	Block the actions of angiotensin II.
Effect on Aldosterone	Reduce aldosterone secretion.	May reduce aldosterone secretion to a lesser extent compared to ACE inhibitors.
Cough Side Effect	More likely to cause a persistent dry cough due to increased bradykinin levels.	Less likely to cause cough compared to ACE inhibitors.
Angioedema Risk	Associated with a higher risk of angioedema, particularly in patients with a history of angioedema.	Lower risk of angioedema compared to ACE inhibitors.
Hyperkalemia Risk	More likely to cause hyperkalemia due to inhibition of aldosterone and potassium retention.	May cause hyperkalemia but generally less than ACE inhibitors.
Renal Protection	Provide renal protection by reducing proteinuria and slowing the progression of chronic kidney disease (CKD).	Similar renal protective effects to ACE inhibitors.
First-line Use in Hypertension	Often considered first-line therapy for hypertension, particularly in patients with diabetes or CKD.	Also considered first-line therapy, especially in patients intolerant to ACE inhibitors due to cough or angioedema.

Erythropoiesis-Stimulating Agents (ESAs)

- Erythropoiesis-Stimulating Agents (ESAs) are medications that *stimulate the production of red blood cells (erythropoiesis) in the bone marrow.*
- **Indications:**
 - Anemia with CKD
 - Anemia hemodialysis
 - those undergoing chemotherapy for cancer.
- **Examples of ESAs:**
 - Epoetin alfa (Epogen, Procrit)
 - Darbepoetin alfa (Aranesp)

- **Mechanism of Action:**
 - ESAs *mimic the action of endogenous erythropoietin* => increased production of red blood cells.
- **Benefits of ESAs in Renal Disease:**
 - **Correction of Anemia**
 - **Improved Quality of Life:** Treatment with ESAs can alleviate symptoms of anemia such as fatigue, weakness, and shortness of breath, improving patients' quality of life.
- **Side effects**
 - Hypertension
 - Thromboembolic Events

Nephrotic syndrome

- **Nephrotic syndrome** is a kidney disorder that causes damage to the glomerular filtration barrier.
- The glomerulus is a network of tiny capillaries in the kidney that filters waste products from the blood.
- In nephrotic syndrome, the damaged glomerulus allows protein to leak into the urine.
 - Hypoalbuminemia
 - High cholesterol
 - increased risk of infection

- Treatment for nephrotic syndrome focuses on two main goals:
 - Reducing proteinuria (protein in the urine)
 - Managing symptoms
- **Corticosteroids – 1st line**
 - work by **reducing inflammation in the glomerulus**, which can help to decrease protein leakage.
 - **Prednisone** is the most commonly used corticosteroid for nephrotic syndrome.
- Diuretics:
- Cholesterol-lowering medications
- Immunosuppressants

Diabetic nephropathy

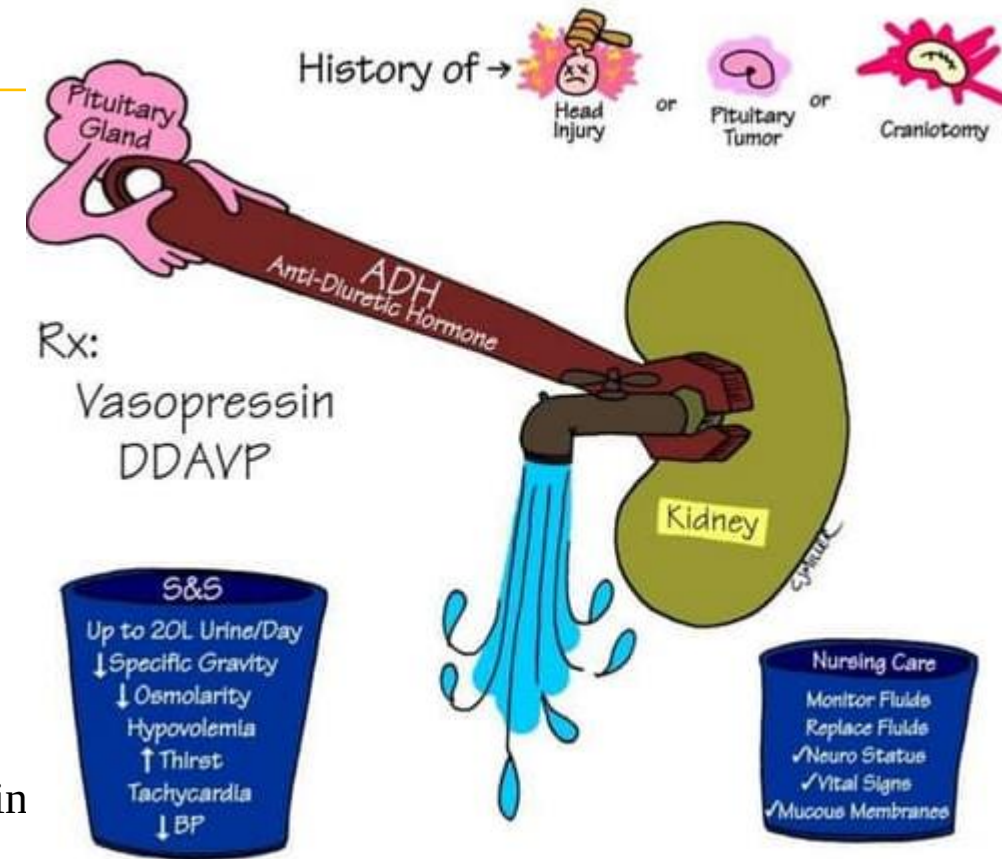
- A complication of diabetes that damages the kidneys' filtration system.
- **Symptoms:**
 - proteinuria,
 - swelling (edema), and
 - decreased kidney function.
- **Risk factors:** Poorly controlled blood sugar, high blood pressure, smoking, and a family history of kidney disease.
- **Complications:**
 - kidney failure,
 - requiring dialysis or a kidney transplant.
 - Also increases the risk of heart disease and stroke.
- **Management:** Focuses on controlling blood sugar, managing blood pressure, and slowing the progression of kidney damage.

Drugs used to manage DN...

- **RAAS inhibitors:** These are the cornerstone of treatment for diabetic nephropathy.
- ACE) inhibitors and angiotensin receptor blockers (ARBs).
- **Sodium-glucose cotransporter 2 (SGLT2) inhibitors:**
 - This is a newer SGLT2 inhibitor work by **lowering blood sugar** levels by causing the kidneys to remove excess glucose from the blood through the urine.
- **Glucagon-like peptide-1 (GLP-1) receptor agonists:**
 - GLP-1 agonists work by **promoting insulin secretion**,
 - Reducing glucagon secretion, and
 - slowing gastric emptying, all of which contribute to better blood sugar control and potentially kidney protection.
 - protecting the kidneys in diabetic patients.
- **Diuretics**
- **Statins:** a class of cholesterol-lowering medications

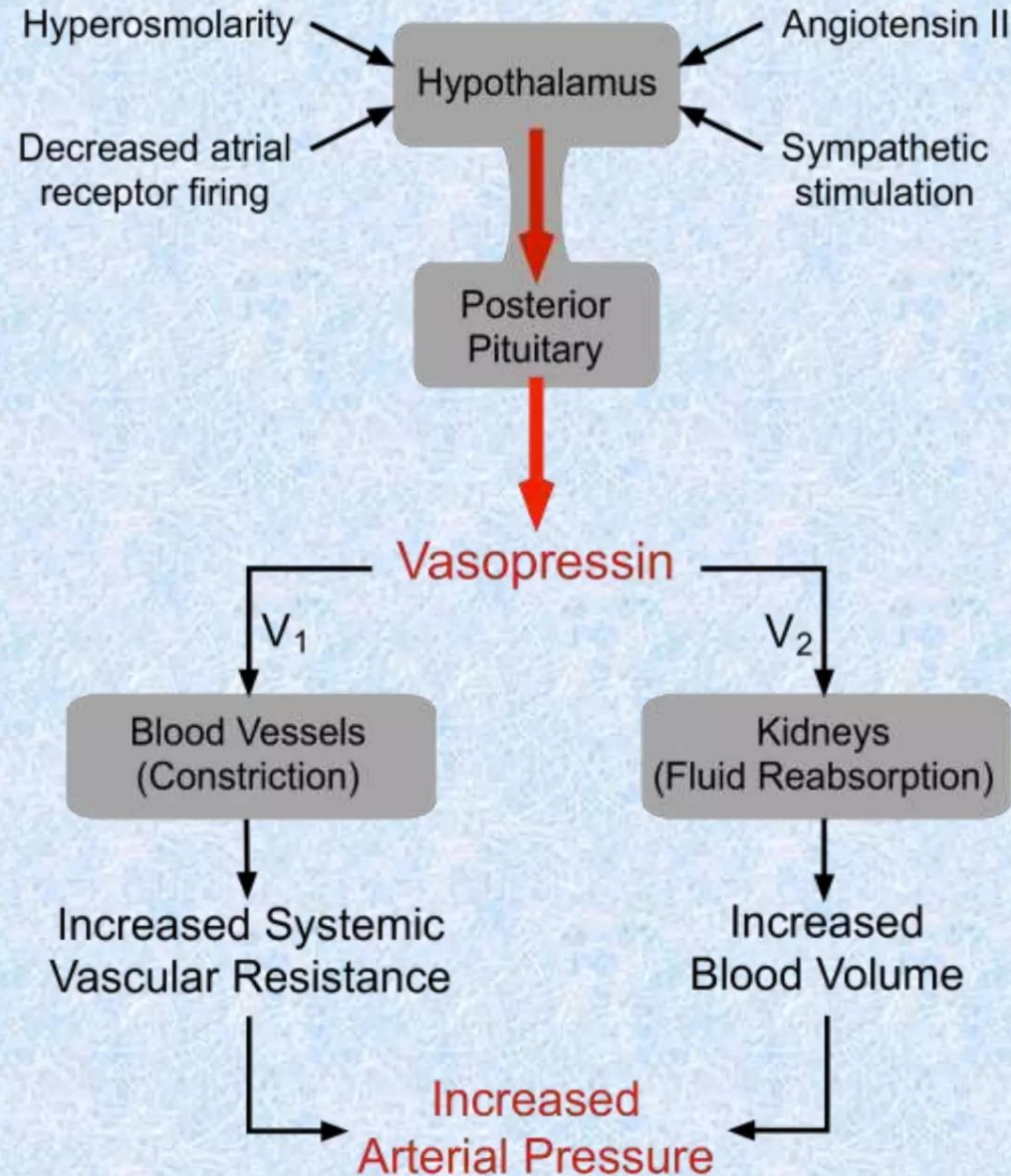
Diabetic Insipidus (DI)

- **Excessive Urine Production:** The primary symptom of DI is **excessive urine production** (polyuria) exceeding normal daily volumes.
- This can lead to dehydration.
- **Underlying Cause:** DI occurs due to a problem with the **hormone vasopressin**, which regulates water reabsorption in the kidneys.
- **Two Main Types:** There are two main types of DI:
 - **Central DI:**
 - Caused by a deficiency in vasopressin production in the hypothalamus.
 - Usually involves a synthetic vasopressin medication to replace the missing hormone
 - **Nephrogenic DI:**
 - Caused by the kidneys' inability to respond properly to vasopressin, even if it's produced normally.
 - May involve medications to manage urine output and address underlying causes if possible.



Antidiuretics

- Vasopressin: it is peptide hormone synthesized in the supraoptic and paraventricular nuclei of the hypothalamus and stored in the posterior pituitary.



Vasopressin analogues

- **Desmopresin:** it is a selective V2 receptors agonists and it more potent than vasopressin as an antidiuretic. It has negligible vasoconstrictor action.
- **Lypressin:** it act both V1 and V2 receptors. It is less potent but longer acting than vasopressin. It is administered parentally.
- **Telipressin:** it is prodrug of vasopressin with selective V1 action. It is administered intravenously.
- **Felypressin:** it is mainly used for its vasoconstrictor (V1) action along with local anesthetics to prolong their duration of action.

Uses of vasopressin analogues

1. Due to V1 receptor-mediated actions:

- For emergency control of bleeding esophageal varices:
Telipressin is preferred over vasopressin because it is safer. It control bleeding by constricting of the mesenteric blood vessels –decrease blood to the portal vessels – reduce pressure in the varices – stops bleeding.
- vasopressin may be used before abdominal radiography to expel intestinal gas.

Cont..

2. Due to V2 receptor- mediated actions:

- Central diabetes insipidus- desmopresin is the drug of choice.
- Hemophilia and von Willebrand's disease- desmopresin, administered intravenously, controls bleeding by increasing factors VIII and von Willebrand's factors.
- Primary nocturnal enuresis: administration of desmopresin at bedtime reduces nocturnal urine volume.

Adverse effect

- Nausea, vomiting, diarrhea, bleeding and abdominal cramps.
- Backache is due to uterine contraction.
- Vasopressin can precipitate an attack of angina by constricting coronary blood vessels. (V1 mediated). Hence, it is contraindicated in patients with hypertension and coronary artery disease.
- Intranasal administration of desmopresin may cause local irritation and ulceration.
- Fluid retention and Hyponatremia can occur (V1 mediated). It should not be given to patients with acute renal failure.

THANK YOU

